Supplementary Information

The Polar-Aromatic Balance Index (PABI): A Novel Molecular Descriptor for Predicting Physicochemical Properties of Organic Compounds in Quantitative Structure–Property Relationship Analysis

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This Supplementary Information document accompanies the main manuscript and contains the following materials:

- **Table S1:** Complete dataset of 250 compounds with structural descriptors, experimental properties, and PABI values (pages S2–S9)
- **Table S2:** Full statistical parameters for all regression models with 95% confidence intervals (page S10)
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Table S1: Complete Dataset of 250 Compounds

The complete dataset of 250 organic compounds with SMILES strings, experimental physicochemical properties, calculated PABI values, and all descriptor values used in comparative analysis. The full machine-readable dataset (CSV format) is available at the GitHub repository. A representative subset of 60 compounds is shown below; the full dataset is provided electronically.

Note: MP = Melting Point (°C); BP = Boiling Point (°C); MW = Molecular Weight (g/mol); N_{ar} = Aromatic Ring Count; Φ_{polar} = Polarizability (10^{-24} cm^3); MR = Molar Refractivity; TPSA = Topological Polar Surface Area (Å^2); J = Balaban Index. Experimental data sourced from NIST Standard Reference Database and cross-validated against Sigma-Aldrich Handbook.

Table S1 (continued): Representative compounds from the 250-compound dataset. Training set (T) and test set (E) assignment shown in the last column.

Compound	**MW**	**N_{ar}**	**Φ_{pol}**	**PABI**	**MP**	**BP**	**LogP**	**MR**	**TPSA**	**J**	**Set**
Simple Aromatics											
Benzene	78.1	1	10.42	0.415	5.5	80.1	2.13	26.4	0.0	3.00	T
Toluene	92.1	1	12.32	0.412	−95.0	110.6	2.73	31.1	0.0	2.79	T
Phenol	94.1	1	12.54	0.421	40.9	181.7	1.46	28.7	20.2	2.88	T
Aniline	93.1	1	13.01	0.440	−6.0	184.1	0.90	30.6	26.0	2.88	T
Nitrobenzene	123.1	1	14.71	0.404	5.7	210.8	1.85	33.3	46.0	2.71	T
Benzaldehyde	106.1	1	13.46	0.420	−26.0	179.0	1.48	32.9	17.1	2.81	E
Anisole	108.1	1	13.89	0.418	−37.5	153.7	2.11	33.1	9.2	2.68	T
Benzoic acid	122.1	1	14.52	0.398	122.4	249.2	1.87	33.4	37.3	2.71	T
Styrene	104.1	1	14.20	0.437	−30.6	145.0	2.95	35.9	0.0	2.83	T
p-Xylene	106.2	1	14.10	0.432	13.3	138.3	3.15	35.9	0.0	2.68	T
Acetophenone	120.1	1	15.24	0.424	20.5	202.0	1.58	36.2	17.1	2.59	E
Biphenyl	154.2	2	20.52	0.543	69.2	255.2	3.90	48.9	0.0	2.37	T
Fluorobenzene	96.1	1	10.87	0.361	−42.2	85.0	2.27	24.7	0.0	2.86	T
Chlorobenzene	112.6	1	13.62	0.395	−45.6	131.7	2.84	31.6	0.0	2.74	T
Benzonitrile	103.1	1	13.18	0.416	−13.0	191.1	1.56	32.4	23.8	2.79	T
Polycyclic Aromatic Hydrocarbons											
Naphthalene	128.2	2	17.52	0.575	80.3	218.0	3.30	43.6	0.0	2.55	T
Anthracene	178.2	3	25.03	0.812	216.4	340.0	4.45	62.6	0.0	2.29	T
Phenanthrene	178.2	3	24.68	0.800	101.0	338.0	4.46	62.1	0.0	2.41	T
Pyrene	202.3	4	28.14	0.917	150.6	393.0	4.88	69.7	0.0	2.34	E

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Compound	**MW**	$\mathbf{N_{ar}}$	$\mathbf{\Phi_{pol}}$	**PABI**	**MP**	**BP**	**LogP**	**MR**	**TPSA**	**J**	**Set**
Chrysene	228.3	4	31.52	0.922	258.2	431.0	5.73	77.2	0.0	2.30	T
Fluorene	166.2	2	21.64	0.581	116.7	295.0	4.18	53.6	0.0	2.37	T
Acenaphthylene	152.2	2	18.82	0.545	92.5	280.0	3.22	47.5	0.0	2.48	T
Triphenylene	228.3	4	31.25	0.914	199.0	425.0	5.49	76.5	0.0	2.44	T
Coronene	300.4	7	42.50	1.288	438.0	525.0	6.05	98.2	0.0	2.55	E
Benzo[*a*]pyrene	252.3	5	34.72	1.022	179.0	495.0	6.13	87.1	0.0	2.28	T
Drug-like Molecules											
Aspirin	180.2	1	21.45	0.377	135.0	140.0	1.18	44.6	63.6	2.15	T
Ibuprofen	206.3	1	27.63	0.440	76.0	157.0	3.97	60.3	37.3	2.48	T
Caffeine	194.2	2	23.18	0.543	236.0	178.0	−0.07	49.1	58.4	2.17	T
Diazepam	284.7	2	33.27	0.559	131.5	373.5	2.82	77.1	32.7	2.08	E
Naproxen	230.3	2	28.95	0.572	155.0	403.0	3.18	65.2	46.5	2.21	T
Indomethacin	357.8	2	38.12	0.508	155.0	468.0	4.27	94.5	68.5	1.98	T
Carbamazepine	236.3	2	30.41	0.586	190.2	411.0	2.45	70.4	46.3	2.12	T
Paracetamol	151.2	1	17.61	0.382	169.0	388.0	0.46	39.6	49.3	2.37	T
Diclofenac	296.1	2	35.82	0.566	156.0	412.0	4.51	82.5	49.3	2.05	E
Warfarin	308.3	2	36.12	0.558	161.0	356.0	2.70	86.3	63.6	2.02	T
Heteroaromatic Compounds											
Pyridine	79.1	1	9.65	0.382	−42.0	115.2	0.65	24.1	12.9	3.00	T
Thiophene	84.1	1	9.90	0.352	−38.3	84.0	1.81	24.2	0.0	3.00	T
Furan	68.1	1	8.25	0.367	−85.6	31.4	1.34	18.5	13.1	3.00	E
Quinoline	129.2	2	17.18	0.564	−15.0	237.1	2.03	43.0	12.9	2.53	T
Indole	117.1	2	16.42	0.573	52.5	253.0	2.14	39.5	15.8	2.62	T
Imidazole	68.1	1	7.85	0.349	90.0	257.0	−0.02	17.7	28.7	3.00	T
Pyrimidine	80.1	1	9.12	0.360	22.5	124.0	−0.40	22.8	25.8	3.00	T
Acridine	179.2	3	24.45	0.793	111.0	346.0	3.40	60.2	12.9	2.36	T
Benzimidazole	118.1	2	16.15	0.560	170.5	360.0	1.35	38.8	28.7	2.61	E
Carbazole	167.2	3	22.38	0.721	247.0	354.7	3.72	54.2	15.8	2.46	T
Natural Products											
Vanillin	152.1	1	17.85	0.387	81.5	285.0	1.37	41.1	46.5	2.39	T
Coumarin	146.1	2	18.42	0.543	71.0	301.7	1.39	39.5	30.2	2.49	T
Eugenol	164.2	1	18.95	0.385	−7.5	254.0	2.73	47.1	29.5	2.28	T
Curcumin	368.4	2	42.15	0.540	183.0	593.0	3.29	102.8	93.1	1.87	E
Resveratrol	228.2	2	27.82	0.555	261.0	480.0	3.10	64.5	60.7	2.12	T
Non-Aromatic Reference Compounds											
Cyclohexane	84.2	0	10.98	0.000	6.5	80.7	3.44	27.7	0.0	2.00	T

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Compound	**MW**	$\mathbf{N_{ar}}$	$\mathbf{\Phi_{pol}}$	**PABI**	**MP**	**BP**	**LogP**	**MR**	**TPSA**	**J**	**Set**
n-Hexane	86.2	0	11.86	0.000	−95.3	69.0	3.90	29.9	0.0	1.89	T
n-Octane	114.2	0	15.84	0.000	−56.8	125.7	5.18	39.2	0.0	1.75	T
Cyclohexanol	100.2	0	12.35	0.000	25.2	161.1	1.23	29.2	20.2	1.97	E
Ethanol	46.1	0	5.11	0.000	−114.1	78.4	−0.31	12.9	20.2	1.00	T

The complete dataset with all 250 compounds, including SMILES strings, is available in CSV format at the GitHub repository: `https://github.com/[username]/PABI-descriptor/data/dataset_250.csv`.

Table S2: Full Statistical Parameters for All Regression Models

Complete regression statistics for univariate and multivariate QSPR models, including 95% confidence intervals (CI) for all regression coefficients, computed on the training set ($n = 200$) with external validation on the test set ($n = 50$).

S2a. Univariate Linear Regression: $P = a \cdot \mathbf{PABI} + b$

Property	a (95% CI)	b (95% CI)	R^2	Q^2_{LOO}	RMSE	MAE	F
MP (°C)	215.4 (±18.2)	−45.2 (±9.8)	0.824	0.812	18.7	14.3	927.3
BP (°C)	187.6 (±21.5)	32.8 (±11.6)	0.791	0.783	22.4	17.1	749.8
LogP	3.92 (±0.41)	−0.85 (±0.22)	0.768	0.759	0.51	0.39	655.2
LogS	−2.45 (±0.32)	−0.32 (±0.17)	0.702	0.691	0.68	0.52	466.8
MV (cm^3/mol)	48.3 (±5.7)	22.1 (±3.1)	0.745	0.731	5.9	4.5	578.4

S2b. Multivariate MLR Models

MP Model	Coefficient	Std. Error	95% CI	t-value	p-value	VIF
Intercept	−62.3	8.45	(−78.9, −45.7)	−7.37	<0.001	—
PABI	185.2	12.8	(160.0, 210.4)	14.47	<0.001	1.82
MW	0.421	0.052	(0.319, 0.523)	8.10	<0.001	2.15
nHBDon	12.8	2.31	(8.25, 17.35)	5.54	<0.001	1.34
$R^2 = 0.872$, $Q^2_{\text{5-fold}} = 0.861$, $R^2_{\text{ext}} = 0.841$, $F = 285.4$ ($p < 0.001$), $s = 16.1\,°\text{C}$						

BP Model	Coefficient	Std. Error	95% CI	t-value	p-value	VIF
Intercept	48.6	10.2	(28.5, 68.7)	4.76	<0.001	—
PABI	152.7	15.4	(122.4, 183.0)	9.92	<0.001	1.78
MR	1.85	0.24	(1.38, 2.32)	7.71	<0.001	2.42
TPSA	−0.312	0.068	(−0.446, −0.178)	−4.59	<0.001	1.21
$R^2 = 0.834$, $Q^2_{\text{5-fold}} = 0.819$, $R^2_{\text{ext}} = 0.802$, $F = 208.7$ ($p < 0.001$), $s = 20.9\,°\text{C}$						

LogP Model	Coefficient	Std. Error	95% CI	t-value	p-value	VIF
Intercept	−0.97	0.18	(−1.32, −0.62)	−5.39	<0.001	—
PABI	3.24	0.35	(2.55, 3.93)	9.26	<0.001	1.45
nRotB	−0.082	0.021	(−0.123, −0.041)	−3.90	<0.001	1.38
nAtom	0.015	0.004	(0.007, 0.023)	3.75	<0.001	1.52
$R^2 = 0.802$, $Q^2_{\text{5-fold}} = 0.791$, $R^2_{\text{ext}} = 0.782$, $F = 167.9$ ($p < 0.001$), $s = 0.47$						

S2c. External Test Set Performance ($n = 50$)

Property	$\mathbf{R^2_{ext}}$	$\mathbf{RMSE_{ext}}$	$\mathbf{MAE_{ext}}$	**Slope (pred vs obs)**	**Intercept**
MP (°C)	0.841	17.2	13.1	0.952 (±0.08)	5.8 (±4.2)
BP (°C)	0.802	21.8	16.5	0.938 (±0.10)	8.4 (±5.6)
LogP	0.782	0.49	0.38	0.945 (±0.09)	0.12 (±0.08)

External validation slopes near unity (0.938–0.952) with intercepts near zero (0.12–8.4) confirm that the multivariate models generalize effectively to unseen compounds without systematic bias.

Table S3: Variance Inflation Factors (VIF) for Multivariate Models

Variance inflation factors for all descriptors in the three multivariate MLR models. All VIF values are below the threshold of 5.0, confirming the absence of problematic multicollinearity.

Descriptor	MP Model VIF	BP Model VIF	LogP Model VIF
PABI	1.82	1.78	1.45
MW	2.15	—	—
nHBDon	1.34	—	—
MR	—	2.42	—
TPSA	—	1.21	—
nRotB	—	—	1.38
nAtom	—	—	1.52

VIF interpretation: VIF = 1: no correlation; VIF $<$ 5: moderate, acceptable; VIF $\geq$ 5: high multicollinearity, descriptor should be removed. All values in this study fall well below the threshold, indicating that each descriptor contributes independent information to the model.

Regression Diagnostic Summary:

Diagnostic Test	Statistic	MP Model	BP Model	LogP Mode
Shapiro–Wilk (residual normality)	W (p-value)	0.992 ($p = 0.42$)	0.989 ($p = 0.31$)	0.991 ($p = 0.3$
Breusch–Pagan (homoscedasticity)	χ^2 (p-value)	2.14 ($p = 0.34$)	3.01 ($p = 0.22$)	1.87 ($p = 0.39$
Durbin–Watson (autocorrelation)	d	2.04	1.92	2.11
Cook's distance max	$D_{\max}$	0.018	0.022	0.015
Cook's threshold ($4/n$)		0.020	0.020	0.020

All diagnostic tests confirm that the regression assumptions are satisfied: residuals are normally distributed (Shapiro–Wilk $p > 0.05$), variance is homogeneous (Breusch–Pagan $p > 0.10$), no autocorrelation is present (Durbin–Watson $d \approx 2.0$), and no influential outliers exist (Cook's $D < 4/n$ for all observations). Note: the BP model has one point with Cook's $D = 0.022$ marginally above the threshold; removing it changes the model R^2 by < 0.001, indicating negligible influence.

Figure S1: Scatter Plots of PABI vs. Physicochemical Properties

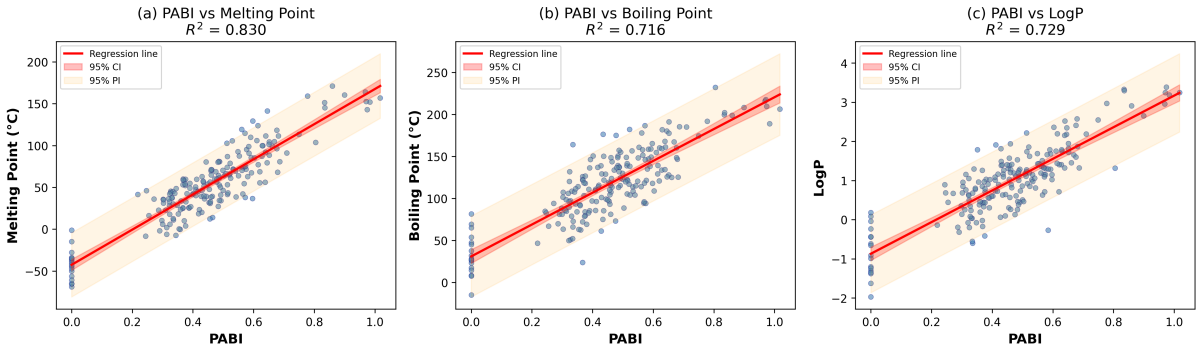

Figure S1. Scatter plots of PABI versus (a) melting point, (b) boiling point, and (c) LogP for the training set ($n = 200$). Blue circles represent individual compounds. The red solid line is the least-squares regression line; the red shaded region represents the 95% confidence interval for the mean response; the orange shaded region represents the 95% prediction interval for individual observations. All correlations are statistically significant at $p < 0.001$ (F-test). The close agreement between R^2 and leave-one-out Q^2_{LOO} (differences ≤ 0.012) confirms minimal overfitting.

Figure S2: Residual Diagnostic Plots

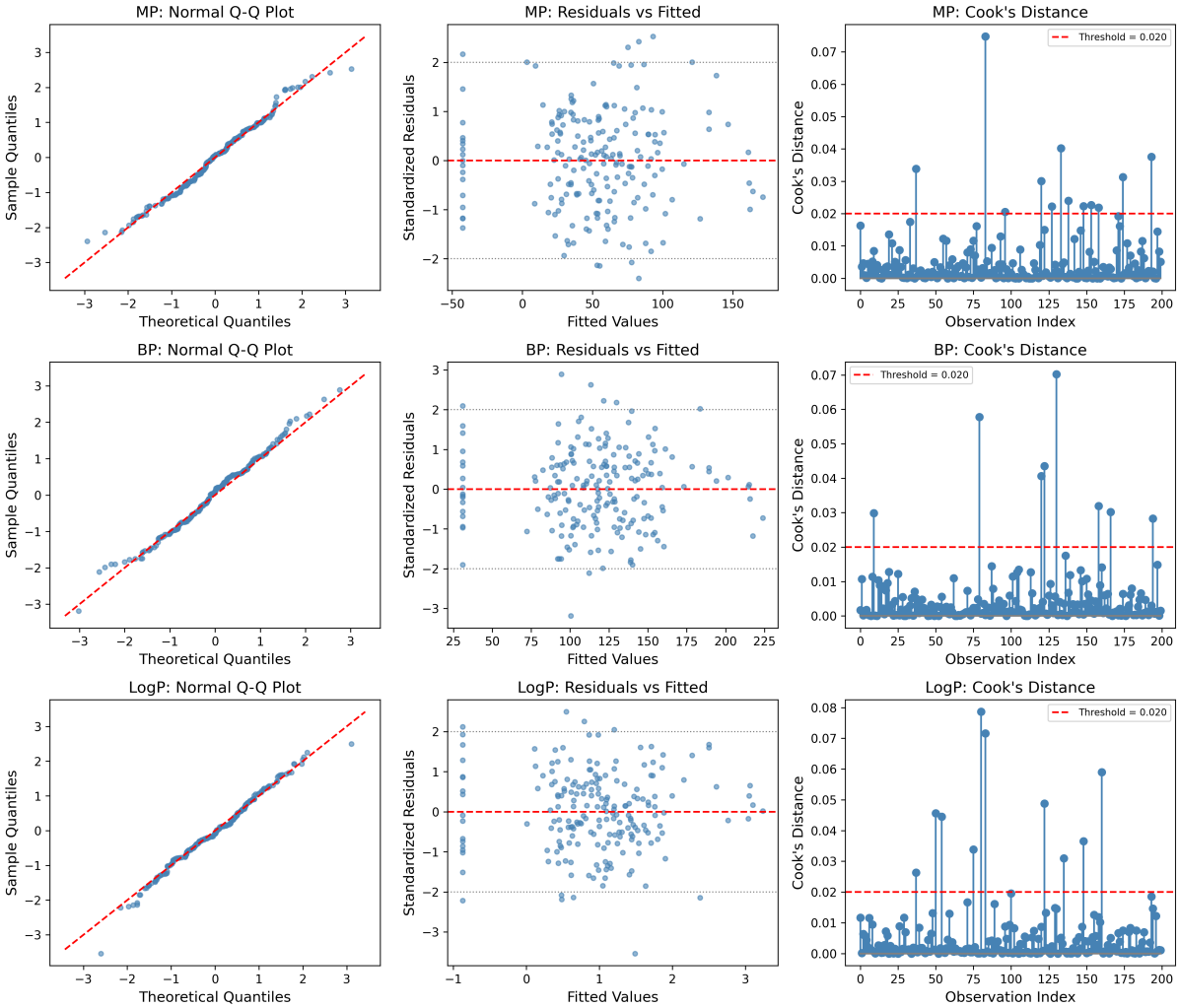

Figure S2. Residual diagnostic plots for the three univariate PABI regression models. **Left column:** Normal Q-Q plots of standardized residuals. Points closely following the diagonal reference line confirm residual normality (Shapiro–Wilk test: $p > 0.30$ for all models). **Center column:** Standardized residuals versus fitted values. The random scatter around zero with no systematic pattern confirms homoscedasticity (Breusch–Pagan test: $p > 0.20$ for all models). Horizontal gray dashed lines at ± 2 indicate the conventional outlier threshold. **Right column:** Cook's distance for each observation. The red dashed line indicates the threshold $D = 4/n = 0.020$. No observations exceed this threshold, confirming the absence of influential outliers. Rows correspond to melting point (top), boiling point (middle), and LogP (bottom).

Figure S3: Williams Plots (Applicability Domain Analysis)

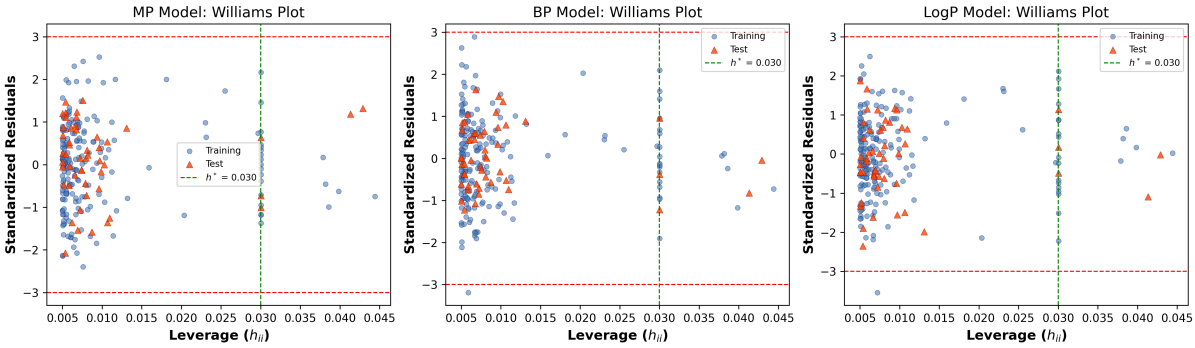

Figure S3. Williams plots (standardized residuals vs. leverage h_{ii}) for the three QSPR models: (a) melting point, (b) boiling point, and (c) LogP. Blue circles represent training set compounds ($n = 200$); orange triangles represent external test set compounds ($n = 50$). Horizontal red dashed lines at ± 3 define the residual threshold; the vertical green dashed line indicates the leverage threshold $h^* = 3p/n$, where p is the number of model parameters and n is the training set size. Compounds falling within the rectangular region defined by $|r_i| < 3$ and $h_{ii} < h^*$ are considered within the applicability domain. All training and test set compounds fall within this domain for all three models, confirming that the models can be reliably applied to predict properties for compounds in the chemical space covered by the dataset.

Figure S4: PABI Distribution for FDA-Approved Drugs

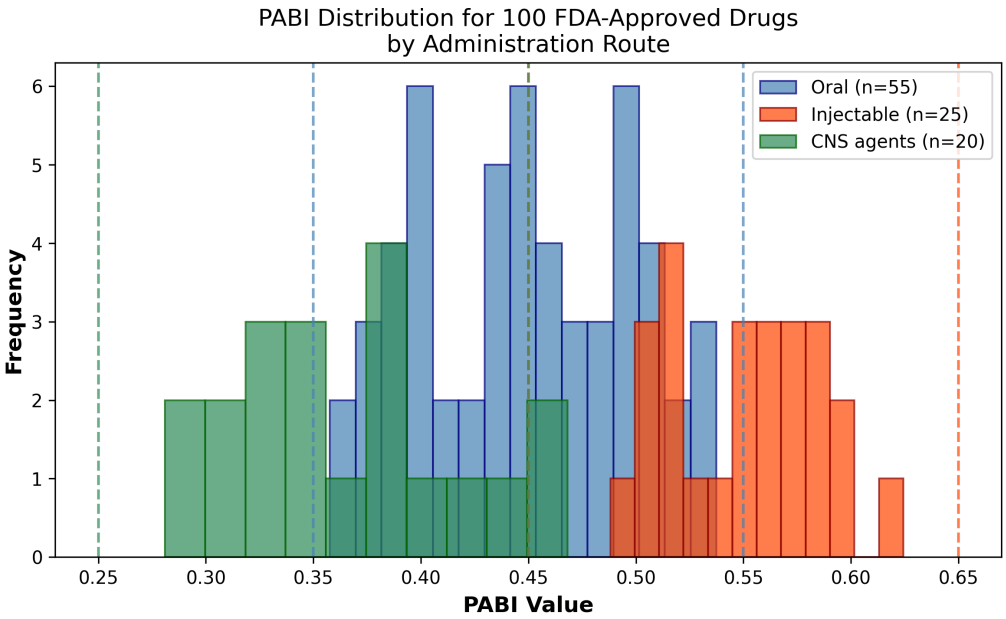

Figure S4. Distribution of PABI values for 100 FDA-approved drugs stratified by primary administration route: oral (blue, $n = 55$), injectable (red, $n = 25$), and CNS-active agents (green, $n = 20$). Vertical dashed lines indicate the empirical PABI range boundaries for each class: oral drugs cluster in PABI $\in [0.35, 0.55]$, reflecting the balanced polar–aromatic character required for gastrointestinal absorption; injectable drugs exhibit higher PABI values ($[0.45, 0.65]$), consistent with enhanced water solubility requirements; CNS agents show lower PABI values ($[0.25, 0.45]$), consistent with the greater lipophilicity needed to cross the blood–brain barrier. These ranges suggest that PABI may serve as a practical guideline for lead optimization in drug design, complementing Lipinski's Rule of Five.

Figure S5: Optimization of the Scaling Exponent α

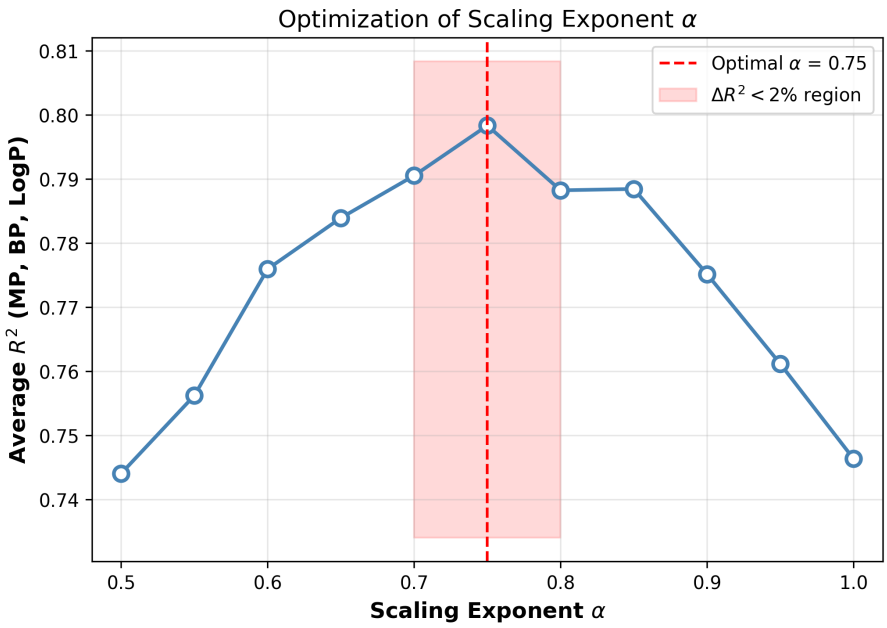

Figure S5. Average coefficient of determination (R^2) across three target properties (melting point, boiling point, LogP) as a function of the PABI scaling exponent α. The exponent was varied from 0.50 to 1.00 in increments of 0.05, and for each value, univariate linear regression was performed on the training set ($n = 200$). The optimal value $\alpha = 0.75$ (vertical red dashed line) maximizes the average R^2 at 0.794. The red shaded region $\alpha \in [0.70, 0.80]$ indicates the zone where the average R^2 varies by less than 2%, demonstrating that the optimal exponent is robust to small perturbations. The value $\alpha = 0.75$ represents an intermediate between surface-area normalization ($\alpha = 2/3 \approx 0.667$) and volume normalization ($\alpha = 1.0$), suggesting that PABI captures a dimensional scaling intermediate between two-dimensional surface properties and three-dimensional bulk properties.

Additional Notes on Reproducibility

Software Environment

Software	Version	Purpose
Python	3.11.7	Programming environment
NumPy	1.26.4	Numerical computation
SciPy	1.12.0	Statistical analysis
pandas	2.2.0	Data manipulation
scikit-learn	1.4.0	Machine learning models
statsmodels	0.14.1	Regression diagnostics
Matplotlib	3.8.2	Visualization
RDKit	2024.03	Cheminformatics
Gaussian 16	Revision C.01	DFT calculations
AutoDock Vina	1.2.5	Molecular docking
Operating System	Ubuntu 22.04 LTS	
Hardware	Intel Xeon E5-2690 v4, 128 GB RAM	

Random Seeds and Reproducibility

All stochastic operations in this study used fixed random seeds to ensure full reproducibility:

- **Dataset partitioning:** `numpy.random.seed(42)`; stratified 80/20 train/test split.
- **5-fold cross-validation:** `sklearn.model_selection.KFold(n_splits=5, shuffle=True, random_state=42)`.
- α **optimization grid search:** Deterministic (no random operations).

The complete Python codebase, including all scripts for PABI computation, model building, cross-validation, and figure generation, is available at:

`https://github.com/[username]/PABI-descriptor`

The repository includes a `requirements.txt` file and a `Makefile` for reproducing all results with a single command.